Data Link Control: Error Detection, Correction & Flow Control

Coverage: Hamming Distance, 2D Parity, CRC-32 Modulo-2 Division, Hamming (7,4) Code, Stop-and-Wait, Go-Back-N, Selective Repeat

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1. Error Control Theory

- Hamming Distance: Detect s errors -> d_min >= s + 1; Correct t errors -> d_min >= 2t + 1.
- CRC: Modulo-2 binary polynomial division using generator polynomial G(x).
- Hamming (7,4) Code: Corrects single-bit errors using syndrome position calculation.

2. Flow Control Protocols

- Stop-and-Wait: Ws=1, Wr=1 | Go-Back-N: Ws=2^m - 1, Wr=1 | Selective Repeat: Ws=2^(m-1), Wr=2^(m-1).

■ Worked Problem: CRC Modulo-2 Division

- Given Data D = 10100100, Generator G = 10101. Remainder R = 1100. Transmitted = 101001001100.

■ KTU University High-Yield Exam Questions

- Q: Explain CRC error detection with a polynomial division example. (8 Marks)
- Q: Derive maximum window sizes for Go-Back-N and Selective Repeat. (8 Marks)